Check for updates

Energy Technology
Generation, Conversion, Storage, Distribution
www.entechnol.de

Boron Minerals with Different Crystal Structures as Performance Manipulators in Triboelectric Nanogenerators

Mehmet Ali Topçu, Abdulkerim Karabiber, Feyyaz Koç, Adem Sarılmaz, and Faruk Özel**

Triboelectric nanogenerators (TENGs) are energy generation devices that naturally convert mechanical movements into electrical energy through both triboelectric and electrostatic effects. The electricity generation capacity of silicone/boron minerals-based composite and glass fiber-based vertical contact separation mode TENGs enhances by adjusting the triboelectrification characteristics by embedding natural boron minerals (NBO) such as borax, colemanite, and ulexite additives. Considering the generated voltages, the NBO-doped silicone and glass fiber-based TENGs offer the maximum performance with 2.5 wt% borax, 7.5 wt% colemanite, and 5 wt% ulexite. The highest power density of corresponding NBO-doped silicone glass fiber TENGs finds as 25.4, 26.03, and 32.03 W m^{-2}, respectively. This work demonstrates that using boron significantly improves the TENG's power and charging capacities fabricated with silicone and glass fiber. Thus, the clean and safe generation of energy will contribute to the reduction of fossil fuel consumption and help protect the environment.

1. Introduction

The world's increasing energy demand is often linked to population growth and the modernization of industrial processes. Still, a majority of the world energy is supplied from fossil fuels however, this leads to environmental problems such as global warming, climate change, desertification, and pollution of air and water sources. Therefore, these environmental problems have directed researchers to emphasize the necessity to switch from conventional resources to alternative green sources, efficient use of energy, and the development of energy conversion devices for sustainable energy production.[1,2] To meet the growing energy demands, different energy harvesting technologies such as solar cells, piezoelectric nanogenerators, thermoelectric nanogenerators, and triboelectric nanogenerators (TENGs) have been developed in recent years.[3,4] Among the all-energy harvesters, TENGs, as energy conversion devices, have gained a lot of attention in recent years, due to being able to harvest electrical energy at both the micro and macro scales with high efficiency.[5] Also, the TENGs can be manufactured from a variety of inexpensive materials to be lightweight and flexible for use in different environments.[6,7]

The TENGs are energy generation devices that convert mechanical energy into electricity by a conjunction of the triboelectric effect and electrostatic induction.[8,9] These novel devices can harvest a wide range of external energies such as body activity,[10,11] air and water stream,[12,13] and sonic waves[14,15] via the piezoelectric and triboelectric effects. Generally, the TENGs are fabricated with two materials with different dielectric properties named positive and negative layers. The important factor for material selection to fabricate the TENG is that the layers that will come into contact with each other have different triboelectric polarities which is a property that states the tendency of a material to gain or lose electrons. There is a very useful chart to select the layers of TENGs, called the triboelectric series, which shows the polarity properties and electron gain/loss trends of a material.[16–18] Up to now, to investigate the output performance of the TENGs, many studies have been reported with various positive materials such as polyamide,[19] poly(vinyl alcohol),[20] different metals,[21,22] ZnO,[23,24] while the negative layer contained polyvinylidene fluoride (PVDF),[25,26] polytetrafluoroethylene (PTFE).[27,28] Also, the triboelectric performance of existing materials can be improved by surface modification and the addition of additive materials such as inorganic/organic nanoparticles, carbon nanotubes, nanofibers, and metals. To improve the TENG output performance Bai et al. manufactured a triboelectric nanocomposite material with nano-Al_2O_3 powders mixed into the cellulose acetate matrix. The nano-Al_2O_3 powder improved the dielectric constant and electrical performance of composite material.[29] Singh and Khare doped ZnO nanorods to PVDF film and fabricated piezo-tribo hybrid nanogenerator by coupling with PTFE to enhance the piezoelectric and triboelectric properties of PVDF. This study showed that the ZnO-PVDF-based generator delivers a remarkable output power of 24.5 μW cm^{-2}, which is 2.5 times higher than the bare PVDF.[30] Yar et al. found that $AgSbS_2$ doped Nylon 6.6 nanofibers improved the both output

M. A. Topçu, A. Sarılmaz, F. Özel
Metallurgical and Materials Engineering
Karamanoğlu Mehmetbey University
Karaman 70200, Türkiye
E-mail: topcumali@kmu.edu.tr; farukozel@kmu.edu.tr

A. Karabiber, F. Koç
Department of Electrical and Electronics Engineering
Bingol University
Bingol 12000, Türkiye

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/ente.202301282.

DOI: 10.1002/ente.202301282

 © 2024 Wiley-VCH GmbH

voltage and power density of TENGs built with polyacrylonitrile nanofibers (PAN).[31] Zhang et al. demonstrated that PVDF nanofibers doped with boron nitride nanosheet exhibit 5 times better triboelectricity than pure PVDF nanofibers.[32] Yar et al. fabricated a TENG with PAN/ZnO and PAN/$(BOH)_3$ nanofibers as tribopositive layers and polyvinyl butyral (PVB) as tribonegative layer. This work showed that the nanogenerator produced with PAN/$(BOH)_3$ and PVB has better triboelectric performance than the PAN/ZnO – PVB nanogenerator.[33] Oh et al. fabricated polydimethylsiloxane – $NiFe_2O_4$ composite as a triboelectric layer for TENG fabrication to determine its electrical properties. Experimental results showed that TENG manufactured with polydimethylsiloxane containing 12 wt% $NiFe_2O_4$ provides 60 V, 300 nA, and 34 nC voltage, current, and charge, respectively.[34] Panda et al. proposed a TENG with biocompatible polydopamine material as humidity sensing layer. The fabricated TENG demonstrates the output voltage and current as 90 V and 2.4 μA, respectively.[35] Nowacki et al. created a TENG via 3D printed method for a detection of ultrasonic waves in water. The study stated that an alternating voltage signal was produced by periodically moving acrylonitrile butadiene styrene pellets by ultrasonic waves.[36]

Recently, the use of silicone-based materials such as silicone elastomer, silicone rubber, and silicone sealant has attracted attention to produce TENG triboelectric layer.[37,38] Silicone-based layers are a suitable material for producing flexible, soft, transparent, waterproof, and lightweight triboelectric layers due to their fluid nature.[39,40] Xiao et al. produced a TENG with silicone rubber/black carbon composite for converting the water wave energy into power. The proposed TENG with maximum power density 2.40 W m^{-3} shows improved output current and voltage.[41] Chen et al. showed that the open circuit voltage produced increases linearly from 3.9 to 17 V with the increasing of the forces for conductive sponge/porous silicone based TENG.[26] Zamanpour et al. investigated the performance of inorganic particle ($BaTiO_3$, Al_2O_3, ZnO, $CaCO_3$, and graphene oxide) doped silicone-based polymer as the charge generating layer. They revealed that the silicone polymer/$BaTiO_3$ composite layer exhibits best performance with 10.10 W m^{-2} power density, 660 V of voltage and 72 μA current.[42] Recently, a new TENG fabricated with silicone sealant as cheap, ease-available, and malleable material, have become attractive. Okbaz et al. investigated Al and Ti-doped silicon sealant as flexible and lightweight material was investigated. As a result of this study, the highest power density of 2.5 wt% Al or 5 wt% Ti embedded silicone glass fiber TENG was found as 22.6 and 21.3 W m^{-2}, respectively.[43] Arkan et al. studied the effect of the dopant structure of CeO_2 on energy harvesting of silicone sealant. This study showed that both CeO_2 nanoparticles and nanofibers at 5 wt%. doping rate provides 80% and 63% times more power than the undoped form.[40] These studies have shown that the doping of additives produced with high-cost and long-term synthesis to different materials has positive effects on TENG performance. In order to produce TENGs in an economical and sustainable way, it is important to use easily available and abundant materials such as silicone sealant and natural resources. In our previous study conducted with pumice-doped silicone sealant, we found that the TENG performance produced with pumice-doped silicone layer and glass giber improved its performance by 36%.[44] So far, no effort has been performed to fabricate TENGs with natural boron (NBO) sources.

The aim of this work is to improve the TENG performance by doping silicon dielectric material with various natural boron minerals. Borate minerals can be found in the earth's crust as various compounds such as colemanite, ulexite, borax, tincalconite, kernite, and kramerite.[45] The boron minerals have been widely used in many areas such as thermal protection, metallurgy, and nuclear industry due to their superior properties such high thermal resistance and electric conductivity.[46–48] They have also found a place in energy practices in the modern world, due to the recently discovered electrical characteristics. In order to use NBO in energy harvesting, TENGs were produced with various ratios of embedded NBO silicone as the negative layer and glass fiber as the positive layer by doctor blade which is an easy and low-cost technique. The output voltage performance was measured with a no-load state. Also, the output voltages were detected under variable loads and different capacitors.

2. Experimental Section

2.1. Fabrication of Nanogenerators and Power Measurements

Silicon dielectric material was supplied commercially from a local market to produce a negative layer of TENGs. Colemanite (C, $2CaO.3B_2O_3.5H_2O$), ulexite (U, $NaCaB_5O_6.5H_2O$), and borax (B, $Na_2B_4O_7.5H_2O$), as NBO was obtained EtiMaden Co., Turkey, and these additives were used without further purification. In order to fabricate the tribonegative layer of nanogenerators, different dielectric materials were produced by the doctor blade technique. Firstly, a mixture was obtained by mixing commercial silicone sealant with 1.0, 2.5, 5.0, 7.5, and 10.0 wt%. powder of NBOs. Then, these mixtures were coated onto an aluminum foil using a template with 10 × 10 × 0.05 cm dimensions by a doctor blade. Finally, the manufactured NBO-doped layers were left overnight to cure and dry (**Figure 1**a).

Vertical contact separation mode TENGs are used to generate electricity as a result of periodic contact and separation of the tribopositive and tribonegative layers. The periodic contact and separation of two triboelectric materials with different properties are provided by a simple mechanism consisting of plexiglass outer layers moving with a spring. The Figure 1b shows the TENG model used in the experiments. In order to test the dielectric of different sizes, the outer layers were cut as 60 × 60 mm. During the application phase, the distance between the lower and upper supporting layers of the TENG was set to 5 mm, taking into account the dielectric and electrode thickness (≈2 mm).

Silicone tribonegative layers doped with NBOs in different proportions were placed on the lower plexiglass layer, and the tribopositive glass fiber layer was placed on the upper layer. Aluminum foil electrodes used as a conductive layer are placed between plexiglass and triboelectric materials. Finally, copper wires were adhered to aluminum foils to allow electric current between the electrodes.

2.2. Instruments and Experiments

X-ray diffraction (XRD) patterns of various NBO are obtained on a Bruker D8 Advance with Cu-Kα radiation source. Surface

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

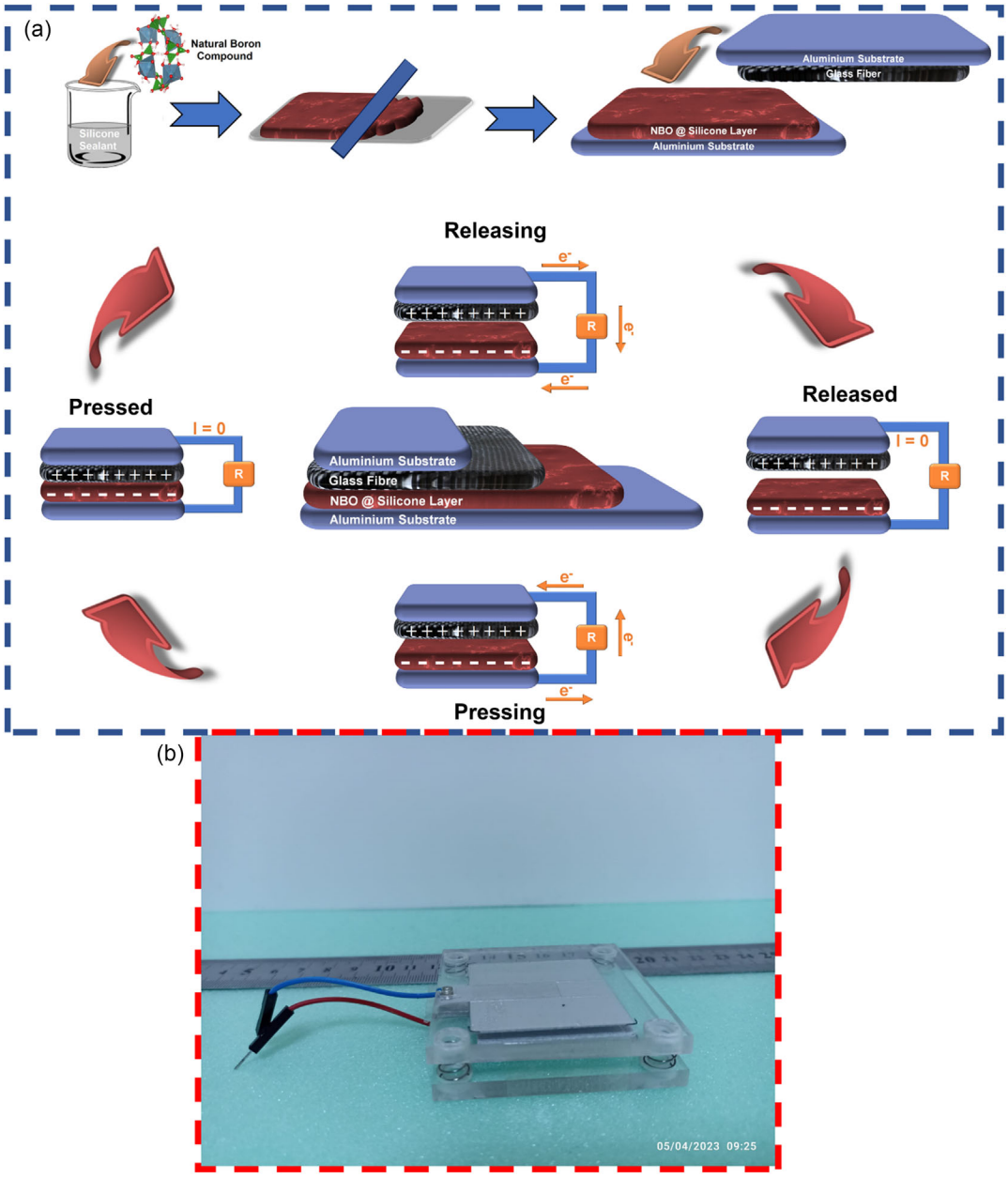


Figure 1. a) Schematic illustration of fabrication process of TENGs and b) digital image of NBO/silicone sealant-based TENG.

morphologies of NBO doped silicone sealant were analyzed with Keyence VHX 900F optical microscope and Zygo ZeGage optical profiler. The output performance of NBO-doped TENGs was determined by a Rigol DS4014 oscilloscope with internal resistance (R_{in}) of 10 MΩ. As stated in previous studies,[43,49] spring-assisted vertical contact-separation mode was used. In the no-load condition, the internal resistance of the measuring instrument ($R_{in} = 10\text{M}\Omega$) is accepted as the load. To obtain larger and smaller loads than R_{in}, external resistors in series and parallel are connected to the TENGs, respectively. Taking into account high force mechanical movements such as the counter force of the spring and stepping movement, the impact force was set as 35 N. A specially designed press machine was used to apply force to TENGs with a controllable movement. The press machine is controlled by programmable logic controller (PLC) and the impact force is measured by Zemīc L6E3 loadcell (Please see Video S1, Supporting Information). During the electrical measurements of fabricated TENGs, in order to charge capacitance and light the LEDs, the AC waveform was converted to a DC waveform using a bridge diode. The maximum output values of TENGs were considered as references for voltage, current, and power curves. Also, the electricity generation application of TENGs by human movement is shown in the supplementary part (Video S2, Supporting Information).

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3. Results and Discussion

3.1. Characterization of Silicon Layer

The XRD patterns of boron compounds doped into silicone layers have been investigated to determine their phase composition and crystal structure. Experimental results are given in **Figure 2**a–d.

As seen from XRD results, boron ores used as additives have different structural properties. Figure 2a represents the XRD pattern of borax. The results indicated that the peak of the XRD assays confirms the purity of the sample. In this structure, the distinctive diffraction peaks indicate that borax has a triclinic structure (JCPDS card no. 00-007-0277). As seen from the XRD analysis of colemanite ore (Figure 2b), the sample contains mainly colemanite in a monoclinic structure (JCPDS card no. 00-033-0267). As stated in previous studies, the colemanite ore may also include some calcite and silica as impurities.[46,50] Figure 2c shows the XRD pattern of ulexite. The results stated that all diffraction peaks belong to ulexite ore. The distinctive diffraction peaks refer that ulexite has a triclinic structure (JCPDS card no. 01-083-1664). The unit cells of these materials are composed of many polyhedras centered on sodium (Na), calcium (Ca), and boron (B). The unit cell of the borax structure consists of 37 polyhedra centered on B and Na, and in this structure, each B atom bonds with 3 or 4 oxygen atoms, while each Na atom makes 6 bonds with O. The colemanite unit cell consists of 16 polyhedra, and unlike borax, the Ca atoms in the structure have 7 bonds with O. The ulexite unit cell, in contrast, consists of 14 polyhedra, and unlike the other two structures, the Ca atoms in the structure have 8 bonds with O (Figure 2d).

The surface morphologies of NBO-doped silicon materials were investigated by optical microscope and optical profilometer, and results are shown in **Figure 3** and **4**. As seen in the optical images (Figure 3), it was observed that the additives were embedded in the silicon layer and formed peaks at some points. This shows that the additives are homogeneously distributed in the silicon layer. Also, it was determined that the surface roughness decreased with increasing additive ratio. Therefore, the S_{pv} (peak to valley height), S_a (arithmetical mean deviation of surface roughness), and S_q (root mean square deviation of surface topography) roughness values are detected with optical profiler (Figure 4). As a result of experimental studies, the S_{pv}, S_q, and S_a values are found as 9.522, 0.890, and 0.727 μm for borax, 11.441, 1.423, and 1.083 μm for colemanite, 28.844, 4.208, and 3.416 μm for ulexite, respectively.

3.2. Power Measurements Results

As a result of the contact of materials with different tribopotentials, opposite loading occurs in the materials. TENGs

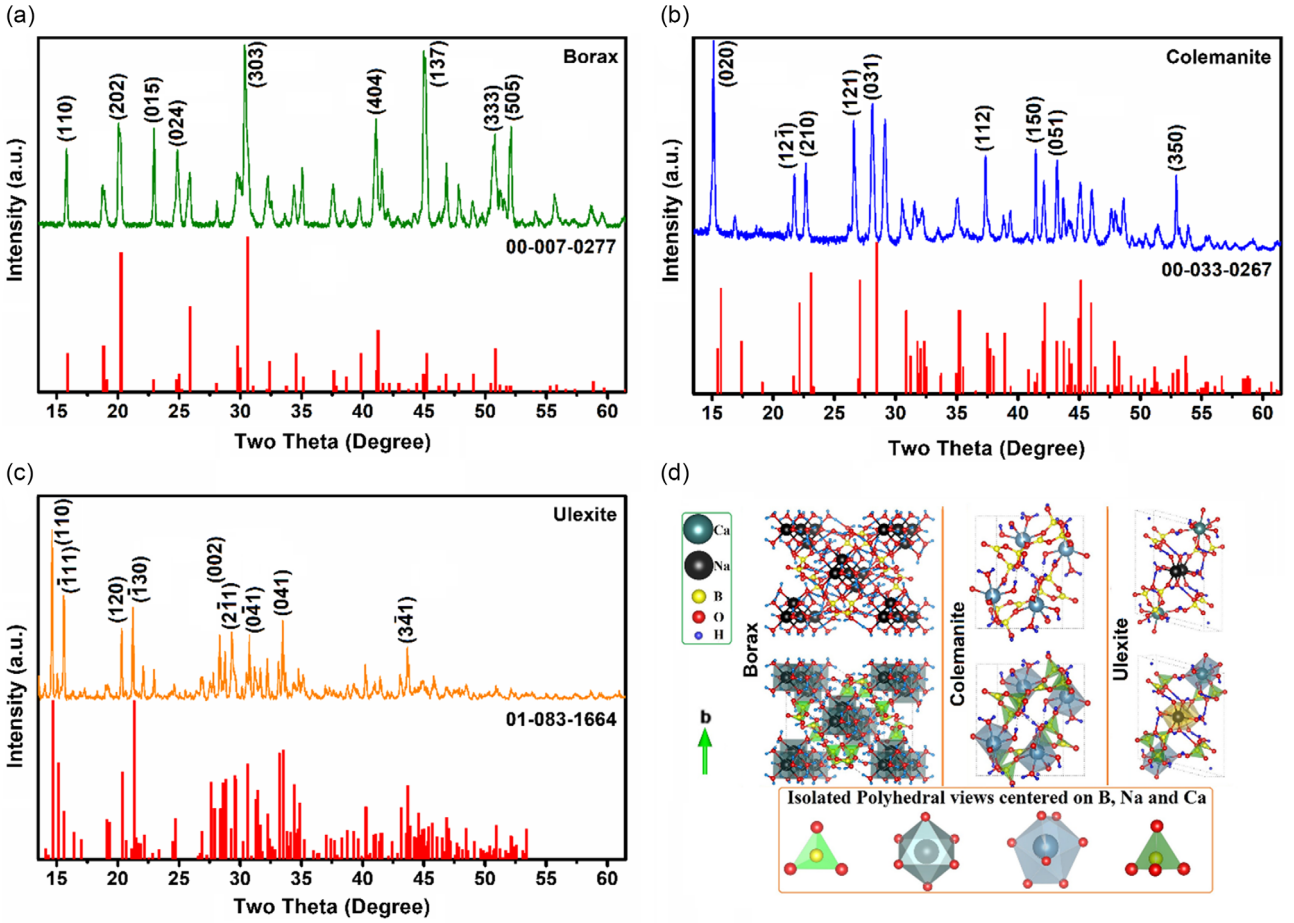


Figure 2. The XRD pattern of boron dopes silicone dielectric materials a) borax, b) colemanite, c) ulexite, and d) corresponding crystal structure.

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

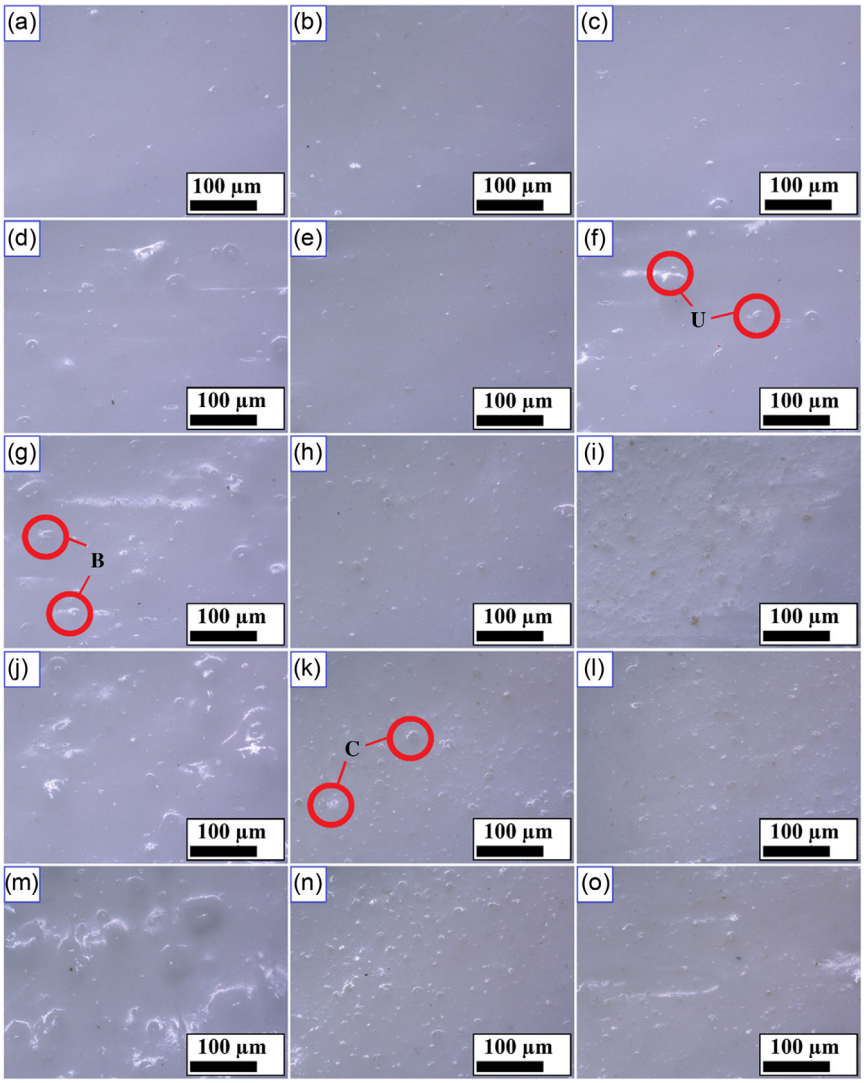


Figure 3. Optical microscope image of borax a,d,g,j,m), colemanite b,e,h,k,n) and ulexite- c,f,i,l,o) doped silicone sealant (1.0% (a–c), 2.5% (d–f), 5.0% (g–i), 7.5% (j–l), 10% (m–o)).

are nanogenerators designed using these properties of materials. In Figure 1, the stages showing the working principle of TENGs are illustrated. While the layers are in contact, one of the dielectric materials gives electrons and the other takes the electrons. After the electron flow, an electrostatic equilibrium occurs between the dielectric materials. Then, while the layers move away from each other, the charge balance is disturbed and electrons pass over a conductive wire to the electrode of the layer. Thus, when the layers move, charge flows occur, and when the layers are stationary, the charge flow stops. In a compression and relaxation cycle, the electron flow occurs in both directions, respectively. Therefore, the electricity produced in TENGs is an AC form and must be rectified before it can be used in electronic devices.

In this study, vertical contact mode TENGs were used to measure the output performance of materials. To measure the voltage output performance of the produced TENGs, the materials were stuck with a force of about 20 N and a frequency range of 2 Hz. **Figure 5** shows the output voltage of TENGs produced with boron minerals doped silicon and glass fiber. To obtain larger and smaller loads than R_{in}, external resistors in series and parallel are connected to the TENGs, respectively. For this reason, both the voltages produced by TENGs in negative and positive alternance are seen in positive alternance.

As seen in Figure 5a, the maximum output voltages of TENGs manufactured with 0, 1, 2.5, 5, 7.5, and 10 wt% of borax-doped silicone sealant are 324.4, 390, 450, 400, 380, and 350 V, respectively. The borax addition to silicone sealant increased the output voltage up to 2.5 wt%. Also, it is understood that TENG produced with 2.5 wt% borax-doped silicone sealant, which has the best output performance, increased the output voltage by 38.9% compared to TENG made with pure silicone sealant. Figure 5b shows the output voltages of TENGs produced with colemanite-doped silicone sealant and glass fiber. The generated maximum voltage

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

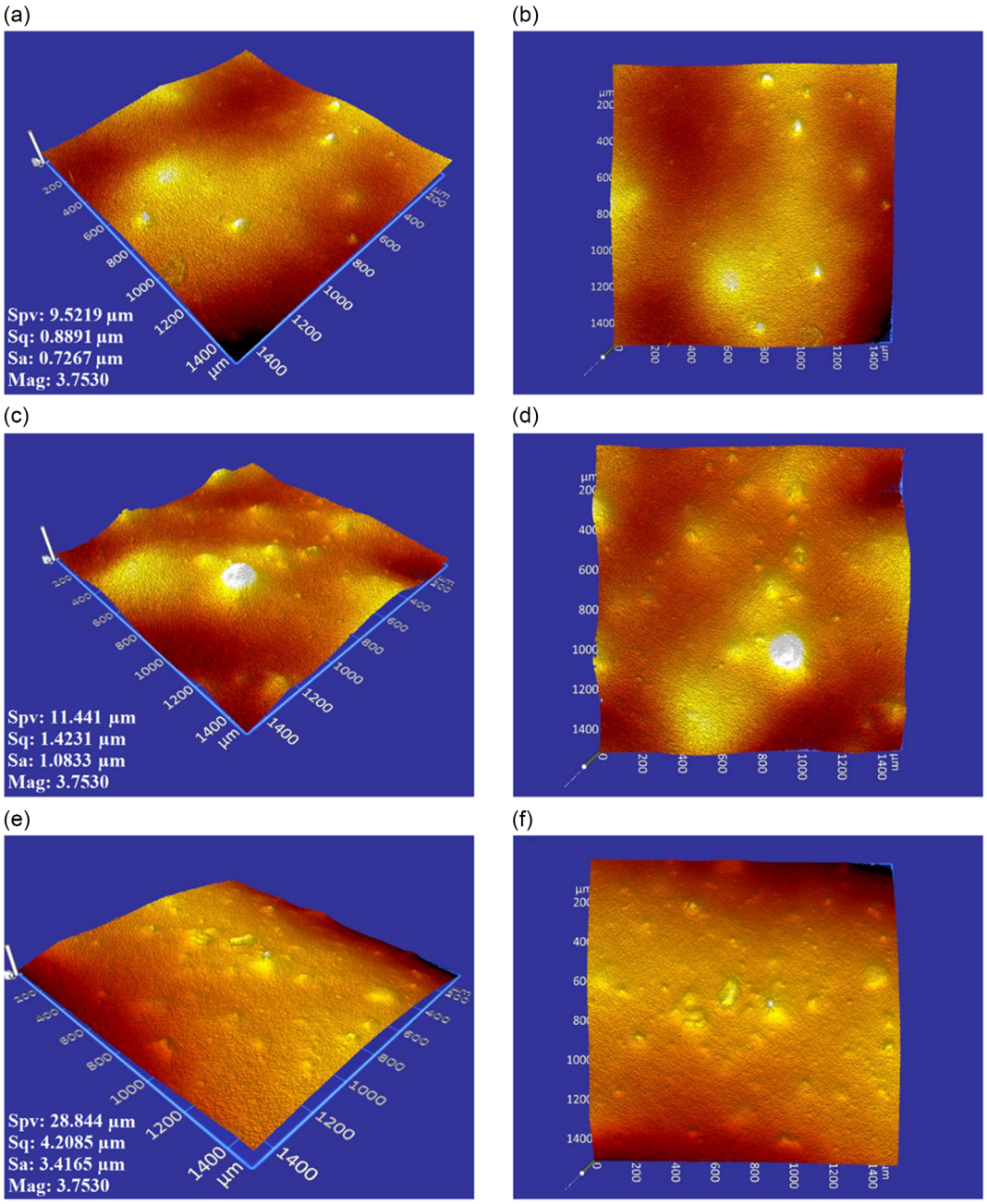


Figure 4. Surface morphology of 2.5 wt% borax a,b), 7.5 wt% colemanite c,d), and 5.0 wt% ulexite- e,f) doped silicon materials.

of TENGs manufactured using 0, 1, 2.5, 5, 7.5, and 10 wt% of colemanite-doped silicone sealant are 324.4, 350, 370, 400, 425, and 354 V, respectively. As a result of power measurements, it was observed that the colemanite addition to the silicone sealant layer increased the output voltage up to 7.5 wt% and then reduced it. In addition, the 7.5 wt% colemanite addition increased the output performance by 31.2% compared to pure silicone sealant. Figure 5c, reveals the output voltage of TENGs produced with ulexite-doped silicone sealant and glass fiber. The generated voltage of TENGs produced with 0, 1, 2.5, 5, 7.5, and 10 wt% of ulexite-doped silicone sealant are 324.4, 450, 490, 505, 430, and 400 V, respectively. The ulexite addition to the silicon provided an increase in output voltage. It is seen that TENG made with 5 wt% ulexite-doped silicone sealant increases the output voltage by 55.9% compared to TENG with pure silicon.

In order to determine the load amount of the produced TENGs and to exhibit the charge-discharge curves, a 0.022 µF capacitor was connected to the TENGs as a load over the bridge diode. Figure 5d shows the capacitor charge-discharge curves of borax-doped TENGs corresponding to one pulse. The measured capacitor voltages of TENGs are 6.25, 7.875, 8.125, 7.5, and 6.5 V,

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

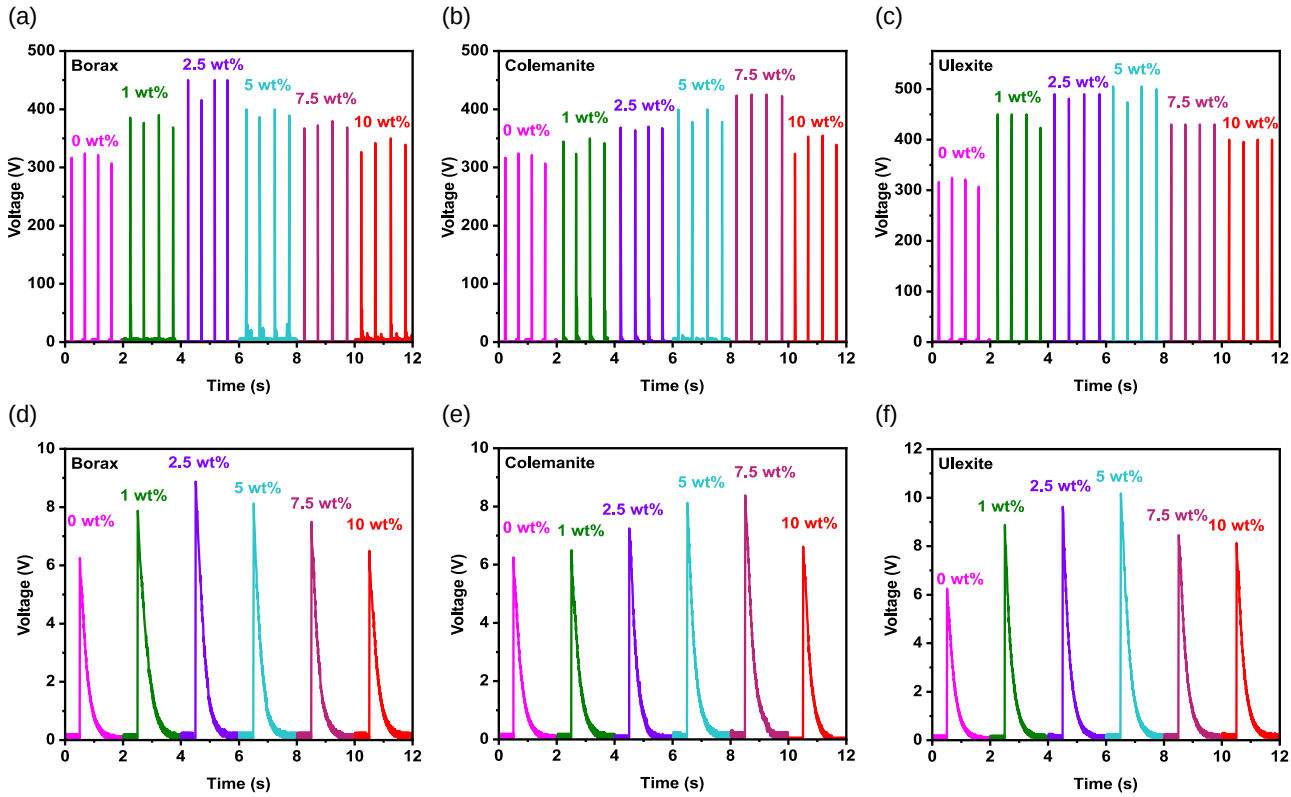


Figure 5. a–c) Output voltage of TENGs for no-load state and d–f) output voltage for a pulse of 0.022 μF capacitive load.

respectively. The charge amounts stored in the capacitor are calculated by the following formula:

$$Q_{\text{stored}} = C{*}V \tag{1}$$

where Q_{stored} is the charge amount stored in the capacitor, C is the capacitance, and V is the voltage produced by TENG.

According to the calculations, the amount of charge in each capacitor was found as 137.5, 172.5, 195.2, 178.75, 165, and 143 nC, respectively. As understood from Figure 5a,d, the output voltages are compatible with the amount of stored charge. It is seen that TENGs with higher output voltages feed the capacitor up to a higher voltage. Figure 5e reveals the charge-discharge curves of colemanite-doped TENGs under a capacitive load of 0.022 μF. The maximum capacitor voltages of colemanite-doped TENGs are 6.25, 6.5, 7.25, 8.125, 8.385, and 6.625 V, respectively. The charge amount stored in the capacitors is found as 137.5, 143, 159.5, 178.75, 184.47, and 145.75 nC, respectively. In Figure 5f, the capacitor charge-discharge curves of ulexite-doped TENGs corresponding to one pulse are illustrated. The measured capacitor voltages of the TENGs produced with ulexite are 6.25, 8.875, 9.625, 10.175, 8.45, and 8.125 V, respectively. The charge amount of in capacitor are calculated with Equation (1) as 137.5, 195.25, 211.75, 223.85, 185.9, and 178.75 nC, respectively.

In **Figure 6**a, power change curves under variable load are given for TENGs produced by silicone sealant with different proportions of borax doped. The maximum power points under 4.7 MΩ load are 17.27, 18.77, 25.4, 19.55, 18.01, and 17.41 mW for 0, 1, 2.5, 5, 7.5, and 10 wt% borax doped silicon, respectively. The power density of 2.5 wt% borax density, which performs the best in terms of power output, is 25.4 W m^{-2}. Figure 6b demonstrates the power change under variable load for TENGs produced with various ratios of colemanite-doped silicone sealant. The maximum points under 4.7 MΩ load are 17.27, 18.77, 20.34, 24.52, 26.03, and 18.47 mW for 0, 1, 2.5, 5, 7.5, and 10 wt% colemanite-doped silicone sealant, respectively. The power density of 7.5 wt% colemanite-doped silicone sealant is 26.03 W m^{-2}. The power changes under variable load for TENGs produced with colemanite-doped silicone sealant are given in Figure 6c. As seen in Figure 6c, the maximum power points under 4.7 MΩ load are 17.27, 26.30, 29.1, 32.03, 24.52, and 22.8 mW for 0, 1, 2.5, 5, 7.5, and 10 wt% colemanite-doped silicone sealant, respectively. The power density of the colemanite-doped silicone sealant with the best performance was determined as 32.03 W m^{-2}. Current and voltage curves under variable load are also given in Figure 6d–f. Since the increase in the load limits the current, the current decreases continuously, but since the increase in the load has a relatively small effect on the voltage, the increase in the voltage continues until the maximum voltage point.

In **Figure 7**a–c, power change curves are shown depending on the contribution ratio. It is seen that the change in the power curves is similar depending on the load and additive ratio. The results show that, borax, colemanite, and ulexite addition increase the tribopotential of silicone sealant, and the best results are obtained at 2.5 wt% additions for borax, 7.5 wt% for colemanite, and 5 wt% for ulexite. Figure 7d shows the voltage curves of

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

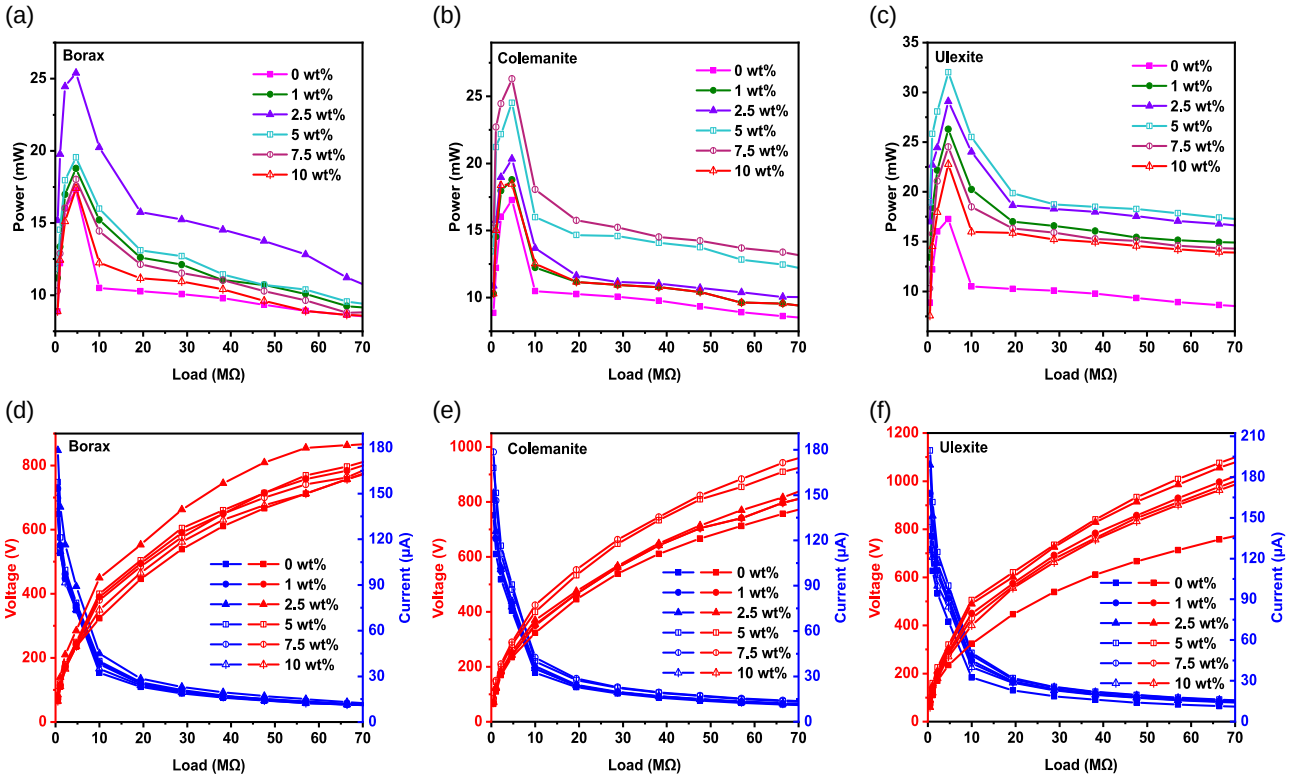


Figure 6. a–c) Maximum power points of TENGs and d–f) current–voltage curves.

TENGs produced with 2.5 wt% borax-doped silicone sealant for different capacitive loads. The capacitor voltages of the TENG for 0.1, 0.3, 0.5, and 1.0 μF loads are 1.59, 0.94, 0.5, and 0.375 V, respectively. As the capacity value increases, the maximum voltage value decreases, and the discharge time increase. In Figure 7e, the voltage values for different capacitive loads of multi-layer TENGs made with colemanite-doped silicone sealant were illustrated. As it is seen, the capacitor voltages of the produced TENGs are 2.5, 1.16, 0.61, and 0.41 V, respectively. Moreover, Figure 7f represents the voltage values for various capacitive loads of multilayer TENGs produced with 5 wt% ulexites-doped silicone sealant. The capacitor voltages of the TENGs are 3.425, 1.65, 0.975, and 0.4825 V respectively. As the capacitance value increases, the maximum voltage value decreases. However, when the amount of stored energy is calculated using Equation (1), it is seen that the amount of stored energy is close to each other. Also, easily available materials such as silicone sealant and glass fiber are known well to be mechanically durable and lightweight materials. In a previous study conducted by our working group, stability tests of glass fiber and silicon-based TENGs were performed under 2 Hz and 100 N force pulses and it was determined that they produced power even after longer than 1000 cycles of use.[44]

The electrical resistivity, dielectric coefficient, and effective surface area of dielectric materials are factors that affect the amount of energy harvested from TENGs. The electrical resistance and capacitance values of the sample silicone sealant were measured to determine the effect of these factors on the output voltage of the TENGs. The measurements were made under a constant force of 35 N, at 200 kHz, and using the two-plate method for $4 \times 4\ cm^2$ samples. **Figure 8**a shows the electrical resistance values of borax, colemanite, and ulexite-added silicones depending on the additive ratio. It is seen that all three samples improve the resistance of silicon as the additive ratio increases. Borax and colemanite are additives that increase electrical resistance the most and least, respectively. According to the output voltage results of TENGs, it is seen that borax-doped TENG is the device that reaches the maximum voltage with the lowest contribution rate (2.5%). This result is due to the fact that the borax additive effectively increases the resistance of silicon. Figure 8b shows the capacitance values of borax, colemanite, and ulexite-added silicones. In all samples, the capacitance values decrease as the additive ratio increases. The highest capacitance value and the fastest decrease are seen in ulexite-added silicones. The effect of borax on the capacitance value of silicon is limited compared to borax and colemanite. The effect of colemanite and ulexite on the capacitance value of silicon is similar and the capacitance values decrease rapidly as the additive ratio increases.

The power losses (P_L) occurring in TENGs are expressed by the following formula:

$$P_L = I_L^2 R \tag{2}$$

where I_L is the leakage current and R is the electrical resistance of the dielectric material. As R increases, the resistance to leakage currents increases and I_L decreases. Since P_L is proportional to the square of I_L, P_L decreases despite the increase in R. With the decrease of P_L, the energy stored in the dielectric material flows

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

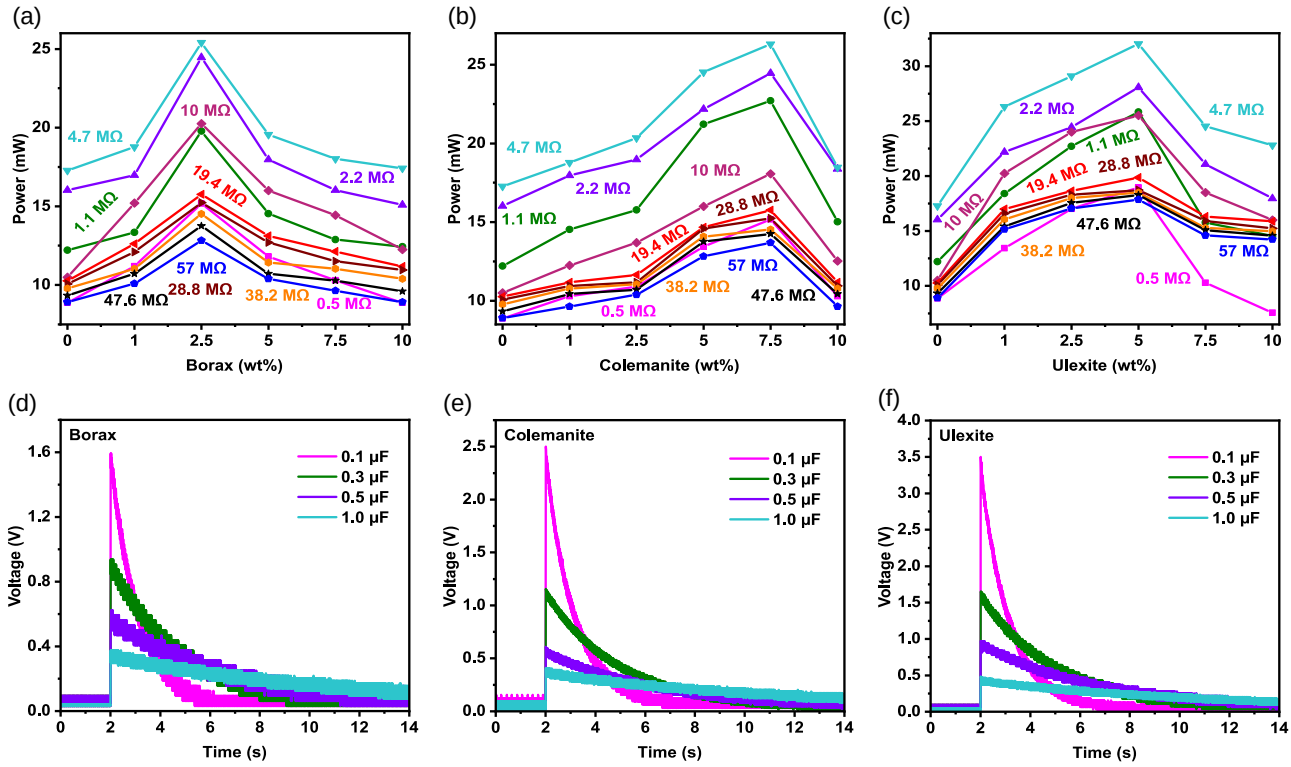


Figure 7. a–c) Maximum output powers depending on the additive ratio and d–f) output voltage corresponding to one pulse for different capacitive loads.

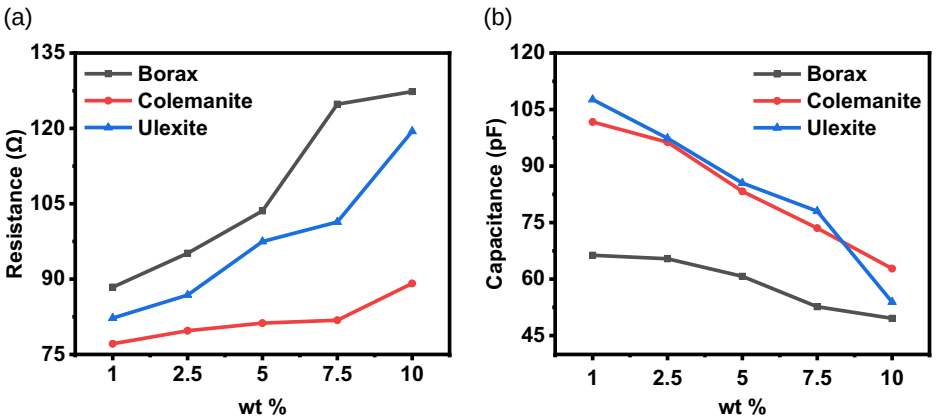


Figure 8. Electrical resistance a) and capacitance b) values of dielectric materials depending on the additive ratio.

through the conductor connected to the plates and increases the output power of TENG. Another important factor affecting the output performance of TENGs is the amount of energy that can be stored in conductive plates. The energy storage capacity (C_T) of TENGs can be found by the following formula:

$$C_T = \frac{\varepsilon A}{d} \tag{3}$$

here, ε is the dielectric constant, A is the effective surface area, and d is the dielectric thickness. The d values are equal in the samples examined in this study. Therefore, as the additive ratio increases, the decrease in C_T indicates that ε or A decreases. The reduction of A is due to irregular protrusions in the surface area of the dielectrics, because these protrusions prevent the dielectric layers from covering each other completely, they cause the formation of air gaps.[51] As the ratio of borax, colemanite, and ulexite additives to silicon increases, R increases, and C_T decreases. The output power of TENGs, which varies proportionally with R and C_T, is crossly affected by this result. Therefore, the output power of TENGs increases up to the optimum additive ratio and then decreases.

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The addition of inorganic bulk or nanoparticles to composites is a method adopted to increase electrical insulation and achieve improvements in dielectric properties.[52–54] Inorganic particle-enhanced composites trigger both interfacial and dipole polarizations in the presence of an external electric field. Interfacial polarization is associated with the movement of charge carriers at the interface between materials. Interactions between the two materials, particularly collisions of charge carriers at the interface, can impede electron movement, thereby increasing resistance. Furthermore, dipole polarization involves the arrangement of dipoles in the material's atoms or molecules under the influence of an electric field. This arrangement can affect the movement of charge carriers, such as electrons. The reorganization of dipoles can restrict electron movement by opposing the force applied by the electric field. This situation can increase resistance by impeding the free movement of charge carriers within the material. The effects of interfacial and dipole polarization can vary depending on the properties of the material and the applied electric field. Careful control of these properties is crucial to optimizing the electrical insulation of inorganic particle-enhanced composites and improving their dielectric performance.

In the last few years, several doping materials have been used in silicone sealant-based TENGs to generate electricity. **Table 1** summarizes the comparison of output performance of NBO doped silicone sealant-based TENGs by other additives.

When the compared to the maximum power density and maximum voltage by ulexite-doped silicone sealant/glass fiber-based TENG with those other additives, the TENG with ulexite have reached relatively higher output performance. Also, borax and colemanite-doped silicone sealant/glass fiber based TENGs revealed higher maximum power density values than Al, Ti, CeO_2 nanoparticle, and pumice added TENGs. But, borax and colemanite has lower maximum voltage. Great efforts have been made in the field of renewable energy with the energy harvesting technologies such as piezoelectricity, thermoelectricity, and solar cells developed to date.[55,56] However, even though the piezoelectric and solar cells provided higher power and energy densities in absolute data, the TENGs may be a better alternative for rechargeable mobile devices due to the costs of production and materials of TENGs are much lower than the costs of piezoelectric materials. Also, triboelectric energy harvesters provide some advantages such as simple construction, easy fabrication, smaller displacement than that required by the piezoelectric energy harvester, and low cost.[57]

Table 1. The output maximum power density and voltage produced by silicone sealant-based TENGs.

TENG	Maximum power density [W m^{-2}]	Maximum voltage [V]	References
Al-doped silicone sealant/glass fiber	22.6	500	[43]
Ti-doped silicone sealant/glass fiber	21.3	490	
CeO_2 nanoparticle-doped silicone sealant/glass fiber	19.32	460	[40]
CeO_2 nanofiber-doped silicone sealant/glass fiber	17.59	445	
Pumice-doped silicone sealant/glass fiber	17.78	430	[44]
Borax-doped silicone sealant/glass fiber	25.40	450	This study results
Colemanite-doped silicone sealant/glass fiber	26.03	425	
Ulexite-doped silicone sealant/glass fiber	32.03	505	

In the light of experimental results mentioned above, it shows that the proposed TENGs is feasible in terms of energy production. The electric generation ability of the new TENGs produced with NBO minerals and easily-available will facilitate the power production for mobile devices. Moreover, NBO doped silicone sealant/glass fiber TENGs, NBO-doped silicone sealant/glass fiber TENGs will support clean energy production in the future.

4. Conclusions

In this study, we aimed to further improve the TENG performance of the silicone sealant dielectric layer known for its superior properties. Since the hybrid materials eliminate each other's adverse effects, the TENG performance of the silicone dielectric layer, which is known to have superior properties, has been improved. In this context, a hybrid dielectric layer composed of different boron ore particles together with silicone material was tested in the TENG. The influence of borax, colemanite, and ulexite conductors embedded at different rates into the silicone sealant layer has been investigated and the optimum additive amount has been determined. While TENG electric outputs for both borax, colemanite, and ulexite augment up to certain values, TENG electrical outputs diminish with a further increase in the embedding rate. The embedding of borax, colemanite, and ulexite considerably increases the surface roughness of the composite triboelectric material. The enhancement in TENG electrical outputs can be ascribed to increased surface roughness, dielectric constant, and transporting and trapping of triboelectric charges into the near surface of triboelectric material. As a result of experimental studies, the highest output voltage has been obtained for 2.5 wt% borax, 7.5 wt% colemanite, and 5 wt% ulexite-doped silicone/glass fiber TENG as 450, 425, and 505 V, respectively. Also, the best output voltage under 0.022 μF was detected as 8.125, 8.385, and 10.175 V for 2.5 wt% borax, 7.5 wt% colemanite, and 5 wt% ulexite-doped silicone sealant/glass fiber TENG. The results showed that the TENG performance was improved by 55.9% by 5 wt% ulexite addition. The fabrication of TENG with boron ore has proven to be a useful method that significantly increases the power and charging capacity of TENGs. Also, due to the water resistance and high stability of silicone sealant, it can be used in many application with different environmental forces.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors thank Dr. Alexander Rowland Lowe and Dr. Emre Arkan for reviewing the manuscript.

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

M.A.T.: Investigation, Formal analysis, Writing - Original Draft; A.K.: Investigation, Validation, Writing - Original Draft, Visualization; F.K.: Investigation, Formal analysis; A.S.: Investigation, Formal analysis; F.O.: Supervision, Conceptualization, Writing - original draft.

Data Availability Statement

Research data are not shared.

Keywords

boron minerals, dielectric properties, energy conversion, silicon, surface properties, triboelectric nanogenerator

Received: October 13, 2023
Revised: December 20, 2023
Published online: January 22, 2024

[1] S. M. A. Z. Shawon, A. X. Sun, V. S. Vega, B. D. Chowdhury, P. Tran, Z. D. Carballo, J. A. Tolentino, J. Li, M. S. Rafaqut, S. Danti, M. J. Uddin, *Nano Energy* **2021**, *82*, 105691.
[2] K. Dong, X. Peng, R. Cheng, Z. L. Wang, *Nanoenergy Adv.* **2022**, *2*, 133.
[3] E. Akman, H. S. Karapınar, *Sol. Energy* **2022**, *234*, 368.
[4] E. Akman, *J. Mol. Liq.* **2020**, *317*, 114223.
[5] K. Dong, Z. L. Wang, *J. Semicond.* **2021**, *42*, 101601.
[6] T. Q. Trung, N. E. Lee, *Adv. Mater.* **2016**, *28*, 4338.
[7] K. Dong, Y. Hu, J. Yang, S.-W. Kim, W. Hu, Z. L. Wang, *MRS Bull.* **2021**, *46*, 512.
[8] M. Li, Y. Jie, L. H. Shao, Y. Guo, X. Cao, N. Wang, Z. L. Wang, *Nano Res.* **2019**, *12*, 1831.
[9] K. Dong, X. Peng, R. Cheng, C. Ning, Y. Jiang, Y. Zhang, Z. L. Wang, *Adv. Mater.* **2022**, *34*, 2109355.
[10] A. Proto, M. Penhaker, S. Conforto, M. Schmid, *Trends Biotechnol.* **2017**, *35*, 610.
[11] R. Zhang, M. Hummelgård, J. Örtegren, M. Olsen, H. Andersson, H. Olin, *Nano Energy* **2019**, *57*, 279.
[12] J. Li, K. Liu, T. Ding, P. Yang, J. Duan, J. Zhou, *Nano Energy* **2019**, *58*, 797.
[13] M. Wang, J. Zhang, Y. Tang, J. Li, B. Zhang, E. Liang, Y. Mao, X. Wang, *ACS Nano* **2018**, *12*, 6156.
[14] S. Cha, S. M. Kim, H. Kim, J. Ku, J. I. Sohn, Y. J. Park, B. G. Song, M. H. Jung, E. K. Lee, B. L. Choi, J. J. Park, Z. L. Wang, J. M. Kim, K. Kim, *Nano Lett.* **2011**, *11*, 5142.
[15] S. N. Cha, J. S. Seo, S. M. Kim, H. J. Kim, Y. J. Park, S. W. Kim, J. M. Kim, *Adv. Mater.* **2010**, *22*, 4726.
[16] Z. L. Wang, *ACS Nano* **2013**, *7*, 9533.
[17] H. Zou, Y. Zhang, L. Guo, P. Wang, X. He, G. Dai, H. Zheng, C. Chen, A. C. Wang, C. Xu, Z. L. Wang, *Nat. Commun.* **2019**, *10*, 1427.
[18] K. Dong, X. Peng, J. An, A. C. Wang, J. Luo, B. Sun, J. Wang, Z. L. Wang, *Nat. Commun.* **2020**, *11*, 2868.
[19] H. Y. Mi, X. Jing, M. A. B. Meador, H. Guo, L. S. Turng, S. Gong, *ACS Appl. Mater. Interfaces* **2018**, *10*, 30596.
[20] H. Patnam, B. Dudem, S. A. Graham, J. S. Yu, *Energy* **2021**, *223*, 120031.
[21] S. Lee, J. I. Hong, C. Xu, M. Lee, D. Kim, L. Lin, W. Hwang, Z. L. Wang, *Adv. Mater.* **2012**, *24*, 4398.
[22] G. Zhu, J. Chen, Y. Liu, P. Bai, Y. S. Zhou, Q. Jing, C. Pan, Z. L. Wang, *Nano Lett.* **2013**, *13*, 2282.
[23] Y. P. Jeon, J. H. Park, T. W. Kim, *Appl. Surf. Sci.* **2018**, *445*, 50.
[24] X. Yang, W. A. Daoud, *Adv. Funct. Mater.* **2016**, *26*, 8194.
[25] T. H. Kong, S.-S. Lee, G. J. Choi, I. K. Park, *ACS Appl. Mater. Interfaces* **2020**, *12*, 17824.
[26] J. Chen, B. Chen, K. Han, W. Tang, Z. L. Wang, *Adv. Mater. Technol.* **2019**, *4*, 1900337.
[27] D. Y. Kim, H. S. Kim, J. H. Jung, *J. Korean Phys. Soc.* **2016**, *69*, 1720.
[28] A. A. R. Mule, B. Dudem, J. S. Yu, *Energy* **2018**, *165*, 677.
[29] Z. Bai, Y. Xu, J. Li, J. Zhu, C. Gao, Y. Zhang, J. Wang, J. Guo, *ACS Appl. Mater. Interfaces* **2020**, *12*, 42880.
[30] H. H. Singh, N. Khare, *Nano Energy* **2018**, *51*, 216.
[31] A. Yar, Z. Kınas, A. Karabiber, A. Ozen, A. Okbaz, F. Ozel, *Renewable Energy* **2021**, *179*, 1781.
[32] J. Zhang, H. Wang, P. Blanloeuil, G. Li, Z. Sha, D. Wang, W. Lei, C. Boyer, Y. Yu, R. Tian, C. H. Wang, *Compos. Commun.* **2020**, *22*, 100535.
[33] A. Yar, A. Karabiber, A. Ozen, F. Ozel, S. Coskun, *Renewable Energy* **2020**, *162*, 1428.
[34] W. Oh, S. Hajra, S. Divya, S. Panda, Y. Oh, Z. Jaglic, P. Pakawanit, T. H. Oh, H. J. Kim, *Mater. Sci. Eng., B* **2023**, *292*, 116397.
[35] S. Panda, H. Jeong, S. Hajra, P. M. Rajaitha, S. Hong, H. J. Kim, *Sens. Actuators, B* **2023**, *394*, 134384.
[36] B. Nowacki, K. Mistewicz, S. Hajra, H. J. Kim, *Ultrasonics* **2023**, *133*, 107045.
[37] Z. Bai, Z. Zhang, J. Li, J. Guo, *Nano Energy* **2019**, *65*, 104012.
[38] R. Zheng, Y. Chen, H. Chi, H. Qiu, H. Xue, H. Bai, *ACS Appl. Mater. Interfaces* **2020**, *12*, 57441.
[39] H. G. Menge, M. W. Kim, S. Lee, Y. T. Park, *ACS Omega* **2023**, *8*, 7135.
[40] E. Arkan, A. Karabiber, M. A. Topçu, Z. Kinas, A. Sarilmaz, S. S. Ozel, F. Ozel, *Surf. Interfaces* **2023**, *37*, 102683.
[41] T. X. Xiao, T. Jiang, J. X. Zhu, X. Liang, L. Xu, J. J. Shao, C. L. Zhang, J. Wang, Z. L. Wang, *ACS Appl. Mater. Interfaces* **2018**, *10*, 3616.
[42] F. Zamanpour, L. Shooshtari, M. Gholami, R. Mohammadpour, P. Sasanpour, N. Taghavinia, *Nano Energy* **2022**, *103*, 107796.
[43] A. Okbaz, A. Karabiber, A. Yar, Z. Kınas, A. Sarılmaz, F. Ozel, *Energy Convers. Manage.* **2022**, *252*, 115053.
[44] M. A. Topçu, A. Karabiber, A. Sarılmaz, F. Koç, F. Özel, *Sustainable Energy Technol. Assess.* **2023**, *57*, 103251.
[45] Y. Kar, N. Şen, A. Demirbaş, *Miner. Energy - Raw Mater. Rep.* **2006**, *20*, 2.
[46] A. Rüşen, *Rev. Rom. Mater.* **2018**, *48*, 245.
[47] A. Rüşen, A. Geveci, Y. A. Topkaya, B. Derin, *Can. Metall. Q.* **2021**, *51*, 157.
[48] A. Rüşen, B. Derin, A. Geveci, Y. A. Topkaya, *JOM* **2016**, *68*, 2316.
[49] Z. Kınas, A. Karabiber, A. Yar, A. Ozen, F. Ozel, M. Ersöz, A. Okbaz, *Energy* **2022**, *239*, 122369.
[50] M. F. Eskibalci, S. G. Ozkan, *Miner. Eng.* **2012**, *31*, 90.
[51] Y. Liu, W. Liu, Z. Wang, W. He, Q. Tang, Y. Xi, X. Wang, H. Guo, C. Hu, *Nat. Commun.* **2020**, *11*, 1599.
[52] M. T. Nazir, B. T. Phung, G. H. Yeoh, G. Yasin, S. Akram, M. S. Bhutta, M. A. Mehmood, S. Hussain, S. Yu, I. Kabir, *Bull. Mater. Sci.* **2020**, *43*, 220.
[53] P. Liu, L. Li, L. Wang, T. Huang, Y. Yao, W. Xu, *J. Alloys Compd.* **2019**, *774*, 396.
[54] Y. Wang, J. Wu, Y. Yin, T. Han, *IEEE Trans. Dielectr. Electr. Insul.* **2020**, *27*, 377.
[55] S. Tangwiwat, S. J. Milne, *J. Met., Mater. Miner.* **2001**, *10*, 28.
[56] S. Saha, R. P. Singh, A. Rout, A. Mishra, A. Ali, H. Basumatary, R. Ranjan, *J. Appl. Phys.* **2023**, *133*, 064101.
[57] P. Thainiramit, P. Yingyong, D. Isarakorn, *Sensors* **2020**, *20*, 5828.

© 2024 Wiley-VCH GmbH

21944296, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ente.202301282 by Universidad Nacional Autonoma De Mexico, Wiley Online Library on [27/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License